VALUATION SCHEME **QPCODE: 25900149**

MAHATMAGANDHIUNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) – MG2DSCPEG100 – DRILLING, TRANSPORTATION AND REFINING TECHNIQUES (2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Part A

Multiple Choice Questions Answer All Questions. Each question carries one mark. (10 × 1= 10)

1. Select which of the following technologies helps to detect leaks and monitor the condition of offshore rigs in real-time. [K] / [CO1]

   A, Drones and remote sensing systems

2. Identify the purpose of casing in an oil well. [U] / [CO1]

   B, To prevent the collapse of the wellbore

3. What environmental risk is associated with improper disposal of oil field water? [K] / [CO1]

   C, Contamination of groundwater

4. Connate water brines are classified as : [U] / [CO1]

   B, Fossil seawater trapped in sediments

5. Which of the following best defines a 'complex refinery'? [U] / [CO3]

   B, A refinery that includes catalytic cracking and reforming units

6. Relate from the following the importance of sweetening process in refinery operations. [U] / [CO4]

   B, To remove sulfur compounds and improve fuel quality

7. A plant requires a heat exchanger for a high-pressure system with temperatures exceeding 260°C. Which exchanger is most suitable? [U] / [CO5]

   C, Shell and tube heat exchanger

8. Identify the natural method of oil recovery. [U] / [CO1]

   C, Dissolved gas drive

9. How would you use the viscosity test to assess the performance of lubricants and oils in practical applications? [K] / [CO6]

   B, The resistance of the fluid to flow

10. The standard method used to determine the cloud point: [U] / [CO6]

    C, ASTM D 2500

Part B

Short Answer Type Questions Answer any four questions. Each question carries three marks. (4 × 3= 12)

11. Outline the difference between offshore and onshore drilling. [K] / [CO1]

Feature	**Onshore Drilling**	**Offshore Drilling**
Location	Land-based drilling	Drilling in oceans or seas
Cost	Relatively lower	High due to specialized equipment
Accessibility	Easier to transport equipment	Requires drilling platforms or rigs

(Give 3 marks to any 2 correct points each for offshore and onshore drilling)

12. Discuss the applications of nitrogen blanketing. [U] / [CO2]

Oil & Gas Industry: Used in the storage of crude oil, gasoline, or other petroleum products to prevent fire hazards, oxidation, and vapor loss. **Food & Beverage Industry:** In tanks where oils, fats, or sensitive beverages are stored, nitrogen blanketing can prevent spoilage and oxidation. **Chemical Industry:** For volatile chemicals, especially solvents, nitrogen blanketing is employed to prevent unwanted chemical reactions, contamination, and loss of product. **Pharmaceutical Industry:** Used in drug manufacturing to protect sensitive compounds and prevent degradation. **(Give full credits to any 3 points)**

13. How does the presence of sulfur analyze the quality and usability of petroleum products? [An] / [CO4]

The presence of sulfur in petroleum products affects quality and usability.

- Sulfur compounds can cause corrosion in engines, pipelines, and refineries.
- Burning sulfur-rich fuels releases sulfur dioxide, leading to air pollution and acid rain.
- High sulfur content requires expensive desulfurization processes to meet regulatory standards.

(Give full credits to 3 points)

14. Describe the differences between LAN (Low Aromatic Naphtha) and HAN (High Aromatic Naphtha) based on composition and applications. [U] / [CO4]

- LAN (Low Aromatic Naphtha) has low levels of aromatic hydrocarbons, while HAN (High Aromatic Naphtha) contains a high percentage of aromatics.
- LAN is mainly used as a feedstock for steam cracking to produce ethylene and propylene, while HAN is used in making solvents, gasoline blending, and petrochemical production.
- LAN is lighter and less toxic, whereas HAN has higher density and better solvency power due to more aromatics. **(Give full credits to 3 points)**

15. Describe how water circulation influence hydrocarbon migration. [U] / [CO1]

Water circulation can significantly influence hydrocarbon migration by:

- Moving hydrocarbons from source rocks towards reservoir rocks by pushing them along pressure gradients.
- Deflecting or redirecting hydrocarbon movement from purely buoyancy-driven vertical migration to more lateral or complex path.
- Strong water flow can either help trap hydrocarbons by sealing them against barriers or cause leakage by flushing hydrocarbons out of reservoirs. **(Any three significant points give full credits)**

16. Illustrate the softening point test of bitumen. [An] / [CO6]

The ring and ball method: The apparatus needed for the test is ring and ball apparatus. The material being tested is heated to (75 – 100℃) above its softening point and stirred until it becomes liquid. The brass rings are heated near to the temperature equivalent to molten material and placed on a metal plate coated with dextrin and glycerin. The test material is placed on the ring and allowed to cool for 30 minutes. Excess material is removed with the help of a knife. Assemble the apparatus with the rings, thermometer and ball guides in position. Fill the bath with distilled water to a height of 50 mm above the upper surface of the rings. The starting temperature should be 5℃. Apply heat to the bath and stir the liquid so that the temperature rises at a uniform rate of 5 ± 0.5℃ per minute. With the increase in temperature of the bath the bitumen material softens up and the steel balls start to sink and carry a portion of material with it. The temperature is noted down when any of the steel balls touches the bottom plate. **(Correct procedure gives 3 marks)**

Part C

Short Essay Type Questions. Answer any four questions. Each question carries seven marks. (7 × 4= 28)

17. Illustrate the environmental concerns associated with extracting petroleum from alternative sources. [A] / [CO1]

Extracting petroleum from alternative sources like oil sands, oil shale, and through coal-to-liquid (CTL) technology can supplement conventional petroleum supplies; they often come with higher costs, environmental concerns, and technical challenges. The future of petroleum production may involve a combination of traditional extraction methods and these alternative sources, depending on technological advancements, economic factors, and environmental considerations. These include high water usage and contamination, land degradation, and significant greenhouse gas emissions during extraction and processing. Additionally, the large-scale production from these sources may contribute to habitat destruction and pollution, further worsening climate change and environmental damage. (**Six points give 7 marks**)

18. Describe the advantages and disadvantages of transporting oil and gas by pipeline, and railway [U] / [CO2]

Advantages of pipelines transporting: Safe and efficient for large volumes, Can operate continuously without the need for drivers or vessels, Cost-effective for long-term transport once infrastructure is in place. Disadvantages: High initial infrastructure costs, Susceptible to leaks or ruptures, leading to environmental damage, Regulatory and land-use challenges, especially for international pipelines.

Advantages of transporting by trains: Can transport large amounts of oil over long distances, Flexible, with ability to access areas without pipelines. Disadvantages: Risk of derailments or accidents, Limited by track infrastructure, especially in remote areas, Potential delays due to congestion or weather conditions.

(3.5 marks each for the correct point for advantages and disadvantages of each transportation method)

19. Give an account on composition of different distillates -Kerosene, Diesel, Fuel oil, Bitumen. [U] / [CO4]

	Kerosene	Diesel	Fuel oil	Bitumen
Boiling Range	140–300 °C	150–370 °C	350–600°C	>500°C
Carbon Chain Range	C9–C17	C9–C20	C22–C44	C >35
Flash point	35 °C	35 °C	66 °C	175 °C
Cetane number		51		
Diesel Index		45		
Smoke point	18 mm			
Pour point		+3 in winter; + 15 summer		
Sulphur content	0.25 (max)	0.25 (max)		
Ductility				75
Softening point				35 °C (min);50 °C (max)
Penetration PEN				80 (min); 100 (max)
Specific gravity				1-0

(Any 2 correct points for each distillates give 1.5 marks and give full credits for 2 points for four distillates)

20. Compare and contrast Natural Rubber Modified Bitumen and Bitumen Emulsion in terms of performance, durability, and environmental impact. [An] / [CO4]

Performance

- NRMB is designed for higher performance in terms of flexibility, elasticity, and resistance to extreme temperatures. The natural rubber modification improves the high-temperature stability and low-temperature flexibility of bitumen, making it suitable for regions with temperature fluctuations and high traffic volumes.
- It offers better resistance to cracking under temperature changes and deformations from heavy traffic, enhancing the overall structural integrity of the road.
- NRMB improves skid resistance, making it safer for driving, especially in wet or icy conditions due to the rubber's grip-enhancing properties.

- Bitumen emulsion is a water-based suspension of bitumen, making it easier to handle and apply at lower temperatures than conventional bitumen. It is generally used for surface treatments and patching work rather than heavy-duty applications like road paving.
- While it performs well for surface dressing (coating aggregates for road surfaces), it does not provide the same long-term structural benefits as NRMB. Bitumen emulsion has lower temperature stability and is not as flexible as NRMB in handling high thermal or traffic stresses.
- Bitumen emulsions are used in lower temperatures (usually between 20°C to 60°C) compared to conventional hot bitumen. They are sprayed onto surfaces or mixed with aggregates.
- The emulsion breaks once applied, with the water evaporating and leaving behind the bitumen to bind the aggregate or surface.

Durability & Environmental Impact

- NRMB is considered more environmentally friendly in some aspects because it often uses recycled rubber from tires, reducing the amount of rubber waste that ends up in landfills. This helps promote sustainability in road construction.
- The durability of NRMB reduces the frequency of road repairs, which in turn decreases the environmental footprint over time due to fewer construction activities and material usage.
- Bitumen emulsion is also considered environmentally friendly because it requires less energy to produce and apply compared to hot bitumen. Since it is water-based, it has lower emissions during the application process, making it safer for workers and the environment.

(Any one point each for performance, durability, and environmental impact of NRMB & BE give 7 marks)

21. Discuss in detail about PMC method of analysis. [U] / [CO6]

PMC is used for determining flash point of fuel above 40℃. In this process sample is directly heated through an electric heater. The flame is introduced at every 1℃ rise in temperature at the surface of liquid. Stirring should be done mechanically by an electric meter. On application of the test flame when a momentary flash obtained id recorded as the flash point of the liquid. Oil under examination is filled up to the mark in the oil cup and then heated by heating the air bath by a burner. Stirrer is worked between tests at a rate of about 1 and 2 revolutions per second. Heat is applied so as to raise the oil temperature by about 5℃ per minute. At every 1℃ rise of temperature, test flame is introduced for a moment by working the shutter. The temperature, at which a distant flash appears inside the cup, is recorded as the flash point. The heating is continued thereafter and its test flame is applied as before. When the oil ignites and continues to burn for at least 5 second, the temperature reading is recorded as the fire point of the oil. **(Correct procedure 7 marks)**

22. Examine the significance of density measurement in industrial applications. [An] / [CO6]

Density is defined as mass per unit volume which helps in determining its overall heaviness or lightness. In the oil and gas industry, understanding how temperature fluctuations can affect the density of hydrocarbons is essential for accurate measurements and calculations during production, transportation, and refining processes. Density measurement is critical as it helps classify crude oils and refined products, which is essential for pricing, transporting, and refining. In refining processes, density measurements are vital for product quality assessment and separation techniques. The understanding of density also aids in transportation logistics, such as pipeline design and material selection, ensuring safe and efficient movement of hydrocarbons. In summary, density is a key property that affects the exploration, extraction, processing, and environmental management of hydrocarbons, making it essential for various industrial applications.

(Any six points with correct explanation give full credit)
